ALGEBRA

बीजगणित

SHEET -02

CLASS ROOM SHEETS / FOR ALL EXAMS

UPDATED

ALGEBRA (बीजगणित)

CLASSROOM SHEET-02

Concept of Perfect Square

1. If $(a-3)^2+(b-4)^2+(c-9)^2=0$, then $\sqrt{a+b+c}$ = ?

(a) –4 (b) 4
(c) ±4 (d) ±2

2. If $(a-1)^2+(b+2)^2+(c+1)^2=0$, then find $2a-3b+7c$ = ?

(a) 12 (b) –11
(c) 3 (d) 1

3. If $(a-4)^2+(b-5)^2+(c-3)^2=0$, then the value of $\frac{a+b}{c}$ is:

(a) 0 (b) 3
(c) 1 (d) –3

4. If $(x+y-z)^2+(y+z-x)^2+(z+x-y)^2=0$, then $x+y+z$ = ?

(a) $\sqrt{3}$ (b) $3\sqrt{3}$
(c) 3 (d) 0

5. If $a^2+b^2+c^2=2(a-b-c)-3$, then the value of $4a-3b+5c$ is –

(a) 2 (b) 3
(c) 5 (d) 6

6. If $a^2+b^2+49c^2+18=2(b-28c-a)$ then the value of (a+b – 7c) is:

SSC CGL 12/04/2022 (Shift- 2)

(a) 4 (b) 3
(c) 2 (d) 1

7. If $a^2+b^2+49c^2+18=2(b-28c-a)$, then the value of (a – b – 7c) is:

SSC CGL 18/04/2022 (Shift- 2)

(a) 2 (b) 1
(c) 3 (d) 4

8. If $a^2+b^2+49c^2+18=2(b+28c-a)$, then the value of (2a – b + 7c) is:

SSC CGL 19/04/2022 (Shift- 2)

(a) – 3 (b) 1
(c) – 4 (d) 5

9. If $a^2+b^2+c^2=2(a+c-1)$, then the value of $a^3+b^3+c^3$ = ?

SSC CGL 12/09/2024 (Shift-03)

(a) 0 (b) 2
(c) 4 (d) 1

QUESTIONS BASED ON $(a+b+c)^2$

10. If $(x+y+z)=23$ and $x^2+y^2+z^2=179$ then find the value of $(xy+yz+zx)$.

SSC Phase-XII 25/06/2024 (Shift-04)

(a) 225 (b) 280
(c) 175 (d) 350

11. If a + b – c = 5 and ab – bc – ac = 10, then find the value of $a^2+b^2+c^2$.

SSC CGL 18/04/2022 (Shift- 01)

(a) 15 (b) 45
(c) 5 (d) 40

12. If $a^2+b^2+c^2=6.25$ and (ab+bc+ca) = 0.52, what is the value of (a+b+c), if (a+b+c)<0?

SSC CGL 11/04/2022 (Shift- 03)

(a) ± 2.7 (b) – 2.7
(c) – 2.8 (d) ± 2.8

13. If $x+y+z=13$, $x^2+y^2+z^2=91$ and $xz=y^2$, then the difference between z and x is:

SSC CHSL 05/08/2021 (Shift- 01)

(a) 3 (b) 8
(c) 5 (d) 9

14. If $a+b+c=3$, $a^2+b^2+c^2=6$ and $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=1$, Here, a, b & c are non-negative, then abc = ?

(a) $\frac{2}{3}$ (b) $\frac{3}{2}$
(c) $\frac{1}{2}$ (d) $\frac{1}{3}$

Questions Based on $(a^3+b^3+c^3-3abc)$

15. If $a+b+c=6$, $a^2+b^2+c^2=14$ and $ab+bc+ca=11$, then what is the value of $a^3+b^3+c^3-3abc$?

SSC CGL 02/12/2022 (Shift-03)

(a) 31 (b) 12
(c) 18 (d) 42

16. If $a+b+c=5$, $a^3+b^3+c^3=85$ and $abc=25$, then find the value of $a^2+b^2+c^2-ab-bc-ca$.

(a) 2 (b) 4
(c) 6 (d) 8

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17. Factorize the given algebraic expression. $x^3 + 27y^3 + 64z^3 - 36xyz$
(a) $(x + 3y + 4z)(x^2 + 9y^2 + 16z^2 + 3xy + 12yz + 4xz)$
(b) $(x + 3y + 4z)(x^2 + 9y^2 + 16z^2 - 12xy^2 - 3yz - 4xz)$
(c) $(x - 3y - 4z)(x^2 + 9y^2 + 16z^2 - 3xy - 12yz - 4xz)$
(d) $(x + 3y + 4z)(x^2 + 9y^2 + 16z^2 - 3xy - 12yz - 4xz)$

18. What is the value of
$$\frac{\left[(2.247)^3 + (1.730)^3 + (1.023)^3 - 3 \times 2.247 \times 1.730 \times 1.023\right]}{\left[(2.247)^2 + (1.730)^2 + (1.023)^2 - 2.247 \times 1.730 - 1.730 \times 1.023 - 2.247 \times 1.023\right]}$$
(a) 1.730 (b) 4
(c) 5 (d) 5.247

19. If a + b + c = 15 and ab + bc + ca = 35, then find the value of $a^3 + b^3 + c^3 - 3abc$.
SSC CPO 29/06/2024 (Shift-01)
(a) 2100 (b) 1500
(c) 1200 (d) 1800

20. If (a + b + c) = 14, and $(a^3 + b^3 + c^3 - 3abc)$ = 98, find the value of (ab + bc + ca).
SSC CGL 26/07/2023 (Shift-04)
(a) 60 (b) 64
(c) 65 (d) 63

21. If (a + b + c) = 12, and $(a^2 + b^2 + c^2)$ = 50, find the value of $(a^3 + b^3 + c^3 - 3abc)$.
SSC CGL 19/07/2023 (Shift-02)
(a) 36 (b) 24
(c) 42 (d) 48

22. If $x + y + z = 2$, $x^3 + y^3 + z^3 - 3xyz = 74$, then $(x^2 + y^2 + z^2)$ is equal to :
SSC CGL 23/08/2021 (Shift- 03)
(a) 22 (b) 29
(c) 26 (d) 24

23. If $x + y + z = 13$, $x^2 + y^2 + z^2 = 133$ and $x^3 + y^3 + z^3 = 847$, then the value of $\sqrt[3]{xyz}$ is:
SSC CPO 24/11/2020 (Shift- 02)
(a) 8 (b) 7
(c) –9 (d) –6

24. If $x = 110$, $y = 111$, $z = 112$, then find the value of $x^3 + y^3 + z^3 - 3xyz$.
SSC CPO 28/06/2024 (Shift-01)
(a) 999 (b) 991
(c) 997 (d) 995

25. If $x = 222$, $y = 223$ and $z = 224$, then find the value of $x^3 + y^3 + z^3 - 3xyz$.
SSC CGL 12/12/2022 (Shift-04)
(a) 2007 (b) 2004
(c) 2006 (d) 2005

26. Find $a^3 + b^3 + c^3 - 3abc$ if $a = 1001$, $b = 1004$ and $c = 1007$.
(a) 80235 (b) 81234
(c) 79356 (d) 81324

27. If $x = z = 225$, and $y = 226$ then $x^3 + y^3 + z^3 - 3xyz$ = ?
(a) 765 (b) 676
(c) 576 (d) 674

28. If a = 2022, b = 2021 and c = 2020, then value of $a^2 + b^2 + c^2 - ab - bc - ca$ is:
SSC CHSL 26/05/2022 (Shift- 03)
(a) 2 (b) 4
(c) 3 (d) 1

29. If p = 106, q = 105 and r = 104, then value of $p^2 + q^2 + r^2 - pq - qr - rp$ is:
SSC Phase-XII 25/06/2024 (Shift-03)
(a) 2 (b) 0
(c) 1 (d) 3

30. If a = 101, b = 102 and c = 103, then $a^2 + b^2 + c^2 - ab - bc - ca$
SSC CGL 24/07/2023 (Shift-01)
(a) 2 (b) 4
(c) 3 (d) 6

31. If $x = 32$, y = 33 and $z = 35$, then evaluate the expression $x^3 + y^3 + z^3 - 3xyz$.
SSC CHSL 01/06/2022 (Shift- 03)
(a) 1120 (b) 1000
(c) 900 (d) 700

32. If $ab + bc + ca = 8$ and $a^2 + b^2 + c^2 = 20$, then possible value of $\frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$ is :
(a) 72 (b) 56
(c) 84 (d) 80

33. If (a + b + c) = 19, and $(a^2 + b^2 + c^2)$ = 155, find the value of $(a - b)^2 + (b - c)^2 + (c - a)^2$.
SSC CGL 18/07/2023 (Shift-02)
(a) 104 (b) 108
(c) 100 (d) 98

34. If $(a^3 + b^3 + c^3 - 3abc)$ = 405, and (a + b + c) = 15, find the value of $(a - b)^2 + (b - c)^2 + (c - a)^2$.
SSC CGL 20/07/2023 (Shift-01)
(a) 27 (b) 54
(c) 18 (d) 45

35. If $(a^3 + b^3 + c^3 - 3abc)$ = 405, and $(a - b)^2 + (b - c)^2 + (c - a)^2$ = 54, find the value of (a + b + c).
SSC CGL 26/07/2023 (Shift-02)
(a) 15 (b) 45
(c) 9 (d) 27

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36. Simplify the following expression.

$$\frac{(59\times59\times59)+(54\times54\times54)+(57\times57\times57)-3(59)(54)(57)}{(59+54+57)}$$

SSC CHSL 27/05/2022 (Shift- 01)

(a) 38 (b) 76
(c) 170 (d) 19

37. If $a = 25$, $b = 15$, $c = -10$, then

$$\frac{a^3+b^3+c^3-3abc}{(a-b)^2+(b-c)^2+(c-a)^2} = ?$$

(a) 30 (b) –15
(c) –30 (d) 15

38. If $a^3 - b^3 - c^3 - 3abc = 0$, then

(a) a = b = c (b) a + b + c = 0
(c) a + c = b (d) a = b + c

39. If a + 2b + 3c = 0, then what is the value of $(a^3 + 8b^3 + 27c^3)$?

SSC CPO 28/06/2024 (Shift-01)

(a) 18 abc (b) 12 abc
(c) 0 (d) 6 abc

40. If a + b = c, then the expression $a^3 + b^3 - c^3 + 3abc$ will be equal to:

SSC CGL 24/09/2024 (Shift-03)

(a) 0 (b) 3
(c) 2 (d) 1

41. If $8a^3 + b^3 + 27c^3 = 18abc$, then the relation among a, b and c is:

SSC CGL 17/09/2024 (Shift-02)

(a) b + 3c = 2a (b) 2a + b = 3c
(c) 2a + 3c = b (d) b + 3c = –2a

42. Simplify the following expression.

$$\frac{(x-y)^3+(y-z)^3+(z-x)^3}{(x-y)(y-z)(z-x)}$$

SSC CGL 13/09/2024 (Shift-01)

(a) 3 (b) 1
(c) 0 (d) 2

43. Find the value of $(1.6)^3 - (0.9)^3 - (0.7)^3$.

(a) 3.24 (b) –3.24
(c) 3.024 (d) –3.024

44. If $(3x + 1)^3 + (x - 3)^3 + (4 - 2x)^3 + 6(3x + 1)(x - 3)(x - 2) = 0$, then what is the value of x?

(a) – 1 (b) $-\frac{1}{2}$
(c) 1 (d) $\frac{1}{2}$

45. If $(4x - 5)^3 + (x - 2)^3 + 27(2x - 5)^3 = 9(4x - 5)(x - 2)(2x - 5)$, then the value of $\left(x+\frac{3}{2}\right)$ will be:

SSC CHSL 05/08/2021 (Shift- 02)

(a) $\frac{1}{2}$ (b) $\frac{5}{2}$
(c) $\frac{3}{2}$ (d) $\frac{7}{2}$

46. If $(5x + 1)^3 + (x - 3)^3 + 8(3x - 4)^3 = 6(5x + 1)(x - 3)(3x - 4)$, then x is equal to:

(a) $\frac{5}{6}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

QUESTIONS BASED ON $(a^3 + b^3)$ and $(a^3 - b^3)$

47. If $a^3 - b^3 = 216$ and $a - b = 6$, then $(a + b)^2 - ab$ is equal to :

(a) 38 (b) 42
(c) 52 (d) 36

48. If a – b = 5 and ab = 25, then the value of $a^3 - b^3$ is:

SSC Phase-XII 25/06/2024 (Shift-02)

(a) 525 (b) 500
(c) 450 (d) 400

49. If (a – b) = 9, and $(a^3 - b^3) = 4410$, find the value of ab.

SSC CHSL 02/08/2023 Shift-02

(a) 190 (b) 112
(c) 162 (d) 136

50. If a + b = 10 and ab = 6, then the value of $a^3 + b^3$ is:

SSC CPO 03/10/2023 (Shift-01)

(a) 860 (b) 820
(c) 800 (d) 840

51. If 3a + 2b = 27 and $27a^3 + 8b^3 = 1458$, then find 2ab.

SSC CGL 18/09/2024 (Shift-03)

(a) 72 (b) 70
(c) 77 (d) 75

52. If $\frac{x}{y}+\frac{y}{x} = 1$ and $x + y = 2$, then the value of $x^3 + y^3$ is:

SSC CPO 03/10/2023 (Shift-01)

(a) 0 (b) 1
(c) 3 (d) 2

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53. If $(8x^3 - 27y^3) \div (2x - 3y) = (Ax^2 + Bxy + Cy^2)$, then the value of (2A + B – C) is :

(a) 4 (b) 6

(c) 5 (d) 3

54. If $2\sqrt{2}x^3 - 3\sqrt{3}y^3 = \left(\sqrt{2}x - \sqrt{3}y\right)(Ax^2 + By^2 + Cxy)$, then the value of $A^2 + B^2 - C^2$ is :

(a) 11 (b) 7

(c) 19 (d) 10

55. If $(8x^3 + 27y^3) \div (2x + 3y) = (Ax^2 + Bxy + Cy^2)$, then the value of (5A + 4B + 3C) is :

(a) 26 (b) 23

(c) 24 (d) 27

56. If $5\sqrt{5}x^3 + 2\sqrt{2}y^3 = \left(Ax + \sqrt{2}y\right)(Bx^2 + 2y^2 + Cxy)$, then the value of $(A^2 + B^2 - C^2)$ is

(a) 15 (b) 20

(c) 30 (d) 40

57. If $27(x + y)^3 - 8(x - y)^3 = (x + 5y)(Ax^2 + By^2 + Cxy)$, then what is the value of (A + B – C)?

CGL 2019 Tier II (15/11/2020)

(a) 13 (b) 16

(c) 18 (d) 11

ANSWER KEY

1.(b)	2.(d)	3.(b)	4.(d)	5.(a)	6.(a)	7.(a)	8.(b)	9.(b)	10.(c)
11.(c)	12.(b)	13.(b)	14.(b)	15.(c)	16.(a)	17.(d)	18.(c)	19.(d)	20.(d)
21.(a)	22.(c)	23.(d)	24.(a)	25.(a)	26.(d)	27.(b)	28.(c)	29.(d)	30.(c)
31.(d)	32.(a)	33.(a)	34.(b)	35.(a)	36.(d)	37.(d)	38.(d)	39.(a)	40.(a)
41.(d)	42.(a)	43.(c)	44.(a)	45.(d)	46.(a)	47.(d)	48.(b)	49.(d)	50.(b)
51.(d)	52.(a)	53.(c)	54.(b)	55.(b)	56.(b)	57.(b)			

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CALCULATION TECHNIQUES

TECHNIQUE – 01



♦ Square Root of perfect square ♦

Calculate the square root of 5184

Step 01: The unit digit of 5184 is 4, so its square root will be end on 2 or 8.

Step 02: Now leave two digits from end and check the remaining number. In this, case it is 51.

$7^2 < 51 < 8^2$

So, the tens digit of square root is 7.

Step 03: Now the square root of 5184 can be 72 or 78. For this, we have to check whether 51 is closer to 7^2 or 8^2 As 51 is closer to 7^2.

Square root of 5184 is 72.



♦ Do it Yourself ♦

(i) $\sqrt{7744}$ = ...

(ii) $\sqrt{1764}$ = ...

(iii) $\sqrt{6561}$ = ...

(iv) $\sqrt{7396}$ = ...

(v) $\sqrt{9216}$ = ...

(vi) $\sqrt{4624}$ = ...

(vii) $\sqrt{7569}$ = ...

TECHNIQUE – 02



♦ Square Root of Non-perfect square ♦

$$\sqrt{N} = \sqrt{(A \pm B)} = \sqrt{A} \pm \frac{B}{2\sqrt{A}}$$

(First find a number which is perfect square near to the number N)

Ex: $\sqrt{10} = \sqrt{9+1} = \sqrt{9} + \frac{1}{2\sqrt{9}} = 3 + \frac{1}{2\times 3} = \frac{19}{6} = 3.17$

Ex: $\sqrt{14} = \sqrt{16-2} = \sqrt{16} - \frac{2}{2\sqrt{16}} = 4 - \frac{2}{2\times 4} = \frac{30}{8} = 3.75$



♦ Do it Yourself ♦

(i) $\sqrt{18}$ = ...

(ii) $\sqrt{29}$ = ...

(iii) $\sqrt{57}$ = ...

(iv) $\sqrt{265}$ = ...

(v) $\sqrt{567}$ = ...

SELF MENTORSHIP TABLE

	STEP-01	STEP-02	STEP-03	STEP-04	STEP-05	STEP-06
TOPIC NAME	VIDEOS	CLASSROOM SHEET	SSC MATHS	NOTES / FORMULA BOOK REVISION	CHAPTER TEST	PRO LEVEL PRACTICE SHEET
ALGEBRA SHEET-02						
CONCEPT OF PERFECT SQUARE						
QUESTIONS BASED ON $(a + b + c)^2$						
QUESTIONS BASED ON $(a^3 + b^3 + c^3 - 3abc)$						
QUESTIONS BASED ON $(a^3 + b^3)$ & $(a^3 - b^3)$						